

# Yingrui Hou

## Data Engineer / Applied Data Scientist | Analytics, Monitoring & Reliable Workflows

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### SUMMARY

Applied data scientist with a PhD in experimental physics and more than three years of postdoctoral experience developing Python/C++ data-processing, statistical-modelling, validation, and monitoring workflows for large, complex datasets.

Experienced in reproducible pipeline development, data-quality diagnostics, ML validation, workflow automation, and root-cause analysis that converted anomalies into calibration or software corrective actions.

Seeking to transfer this technical foundation to reliable digital-agriculture services and analytical products while developing advanced R, DevOps/MLOps, and platform-operations expertise.

### CORE SKILLS

- Data processing: Python, NumPy, pandas, large-scale analysis workflows, structured data inputs and outputs
- Machine learning: scikit-learn, statsmodels, CatBoost / Gradient Boosting, feature studies, train/test validation
- Monitoring and quality: data validation, anomaly diagnosis, expected-versus-observed checks, root-cause analysis
- Workflow engineering: C/C++, Linux, YAML / Shell / CMake automation, reusable pipeline components
- Software practices: Git, CI/CD, shared codebases, technical review, reproducibility, documentation
- Analytics: statistical modelling, time-series analysis, calibration, uncertainty quantification, matplotlib, seaborn

### PROFESSIONAL EXPERIENCE

#### Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Developed reusable Python/C++/ROOT data-processing and analysis workflows for large, noisy datasets, replacing one-off analyses with structured and reproducible pipelines.
- Built monitoring workflows that compared expected and observed behaviour, identified data or software anomalies, and supported calibration updates and corrective actions.
- Developed CatBoost-based pipelines with feature studies, train/test diagnostics, threshold scans, saved model outputs, and reproducible application across datasets.
- Automated data preparation, statistical extraction, validation checks, and reporting inputs using Python/C++, YAML, Shell, and shared configuration.
- Integrated simulated, control, and measured datasets through reweighting, calibration, and data-quality checks before downstream analysis and decisions.
- Worked in Linux-based shared codebases using Git and CI/CD practices, with collaborative review, technical documentation, and reproducible delivery.
- Diagnosed concrete detector and software issues, including a missing-data problem later corrected in the reconstruction workflow, rather than stopping at descriptive monitoring.

#### Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed simulation and data-analysis software combining C++, automated processing, calibration, performance evaluation, and validation against experimental measurements.

## SELECTED OUTCOMES

- Built validation workflows that exposed actionable data and software issues and supported downstream corrective actions.
- Improved reconstructed-data resolution by about 2x through time-resolved analysis, model correction, and calibration on realistic measurements.

## SELECTED PROJECTS

### Test-Data Validation & Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

*Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics*

### Domain Adaptation for ML

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

*Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting*

### Weak-Signal Inference

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

*Methods: likelihood fitting, control-sample validation, production yield workflows*

### Model Correction Workflow

Developed simulation-correction tooling that turns measured physics constraints into an event-level resampling workflow.

*Methods: Gaudi-package development, event-level correction, framework integration*

### Time-Series Inference

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

*Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting*

## EDUCATION

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2